SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network block.

Department	Subnet Mask	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
School of Computing	/25 (255.255.255.128)	**192.168.100.0**	192.168.100.1	192.168.100.126	**192.168.100.127**	**126**
School of Electronics	/26 (255.255.255.192)	**192.168.100.128**	192.168.100.129	192.168.100.190	**192.168.100.191**	**62**
School of Mechanical Engineering	/27 (255.255.255.224)	**192.168.100.192**	192.168.100.193	192.168.100.222	**192.168.100.223**	**30**

Calculations

1. School of Computing (/25)

- **Subnet Mask:** 255.255.255.128
- **Block Size:** 128 addresses
- **Network Address:** 192.168.100.0
- **Broadcast Address:** 192.168.100.127
- **Valid Host Range:** 192.168.100.1 – 192.168.100.126
- **Usable Hosts:** $2^{32-25} - 2 = 128 - 2 = 126$

2. School of Electronics (/26)

- **Subnet Mask:** 255.255.255.192
- **Block Size:** 64 addresses
- **Network Address:** 192.168.100.128
- **Broadcast Address:** 192.168.100.191
- **Valid Host Range:** 192.168.100.129 – 192.168.100.190
- **Usable Hosts:** $2^{32-26} - 2 = 64 - 2 = 62$

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3. School of Mechanical Engineering (/27)

- **Subnet Mask:** 255.255.255.224
- **Block Size:** 32 addresses
- **Network Address:** 192.168.100.192
- **Broadcast Address:** 192.168.100.223
- **Valid Host Range:** 192.168.100.193 – 192.168.100.222
- **Usable Hosts:** $2^{32-27} - 2 = 32 - 2 = 30$

Remaining Unused IP Address Range

After allocating the three subnets:

- Computing: **192.168.100.0 – 192.168.100.127**
- Electronics: **192.168.100.128 – 192.168.100.191**
- Mechanical: **192.168.100.192 – 192.168.100.223**

The remaining unused portion of the original **192.168.100.0/24** network is:

- **Unused Network Range: 192.168.100.224 – 192.168.100.255**
- This corresponds to a **/27 subnet**.
- **Network Address:** 192.168.100.224
- **Broadcast Address:** 192.168.100.255
- **Valid Host Range:** 192.168.100.225 – 192.168.100.254
- **Usable Hosts Available: 30**

Final Answer (Summary)

Department	Network Address	Broadcast Address	Host Range	Usable Hosts
School of Computing	192.168.100.0/25	192.168.100.127	192.168.100.1 – 192.168.100.126	126
School of Electronics	192.168.100.128/26	192.168.100.191	192.168.100.129 – 192.168.100.190	62
School of Mechanical Engineering	192.168.100.192/27	192.168.100.223	192.168.100.193 – 192.168.100.222	30
Unused Range	**192.168.100.224/27**	**192.168.100.255**	**192.168.100.225 – 192.168.100.254**	**30**

Solution

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
School of Computing	/25 (255.255.255.128)	**192.168.100.0**	**192.168.100.1 – 192.168.100.126**	**192.168.100.127**	**126**
School of Electronics	/26 (255.255.255.192)	**192.168.100.128**	**192.168.100.129 – 192.168.100.190**	**192.168.100.191**	**62**
School of Mechanical Engineering	/27 (255.255.255.224)	**192.168.100.192**	**192.168.100.193 – 192.168.100.222**	**192.168.100.223**	**30**

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1. Network Address and Broadcast Address

School of Computing (/25)

- Network Address: **192.168.100.0**
- Broadcast Address: **192.168.100.127**

School of Electronics (/26)

- Network Address: **192.168.100.128**
- Broadcast Address: **192.168.100.191**

School of Mechanical Engineering (/27)

- Network Address: **192.168.100.192**

- Broadcast Address: **192.168.100.223**

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2. Valid Host IP Address Range

School of Computing

- First Host: **192.168.100.1**
- Last Host: **192.168.100.126**

School of Electronics

- First Host: **192.168.100.129**
- Last Host: **192.168.100.190**

School of Mechanical Engineering

- First Host: **192.168.100.193**
- Last Host: **192.168.100.222**

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3. Number of Usable Host Addresses

The formula is:

Usable Hosts = 2^(32 − Prefix Length) − 2

School of Computing (/25)

- Total addresses = 2^(32−25) = 128
- Usable hosts = **126**

School of Electronics (/26)

- Total addresses = 64
- Usable hosts = **62**

School of Mechanical Engineering (/27)

- Total addresses = 32
- Usable hosts = **30**

4. Remaining Unused IP Address Range

Allocated subnets occupy:

- **192.168.100.0 – 192.168.100.127** (/25)
- **192.168.100.128 – 192.168.100.191** (/26)
- **192.168.100.192 – 192.168.100.223** (/27)

The remaining unallocated addresses are:

Unused Range	Total Addresses
192.168.100.224 – 192.168.100.255	**32 addresses**

This unused range corresponds to the subnet:

- **Network Address:** 192.168.100.224/27
- **Broadcast Address:** 192.168.100.255
- **Valid Host Range:** 192.168.100.225 – 192.168.100.254
- **Usable Hosts:** 30

Final Answer (Summary)

Department	Subnet	Network Address	Host Range	Broadcast Address	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	/26	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62

Remaining Unused IP Range

- **Unused Network:** 192.168.100.224/27

- **Available IP Range: 192.168.100.224 – 192.168.100.255**
- **Usable Host Range: 192.168.100.225 – 192.168.100.254**
- **Broadcast Address: 192.168.100.255**
- **Usable Hosts Available: 30**

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

Solution

Department	Subnet Mask	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
Software Development	/24 (255.255.255.0)	**10.10.0.0**	10.10.0.1	10.10.0.254	**10.10.0.255**	**254**
Quality Assurance	/25 (255.255.255.128)	**10.10.1.0**	10.10.1.1	10.10.1.126	**10.10.1.127**	**126**
Human Resources	/26 (255.255.255.192)	**10.10.1.128**	10.10.1.129	10.10.1.190	**10.10.1.191**	**62**
Executive Management	/27 (255.255.255.224)	**10.10.1.192**	10.10.1.193	10.10.1.222	**10.10.1.223**	**30**

Calculations

1. Software Development (/24)

- **Subnet Mask:** 255.255.255.0
- **Network Address:** 10.10.0.0
- **Broadcast Address:** 10.10.0.255
- **Valid Host Range:** 10.10.0.1 – 10.10.0.254
- **Usable Hosts:** $2^{32-24} - 2 = 256 - 2 = 254$

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2. Quality Assurance (/25)

- **Subnet Mask:** 255.255.255.128
- **Network Address:** 10.10.1.0
- **Broadcast Address:** 10.10.1.127
- **Valid Host Range:** 10.10.1.1 – 10.10.1.126
- **Usable Hosts:** $2^{32-25} - 2 = 128 - 2 = 126$

3. Human Resources (/26)

- **Subnet Mask:** 255.255.255.192
- **Network Address:** 10.10.1.128
- **Broadcast Address:** 10.10.1.191
- **Valid Host Range:** 10.10.1.129 – 10.10.1.190
- **Usable Hosts:** $2^{32-26} - 2 = 64 - 2 = 62$

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4. Executive Management (/27)

- **Subnet Mask:** 255.255.255.224
- **Network Address:** 10.10.1.192
- **Broadcast Address:** 10.10.1.223
- **Valid Host Range:** 10.10.1.193 – 10.10.1.222
- **Usable Hosts:** $2^{32-27} - 2 = 32 - 2 = 30$

Remaining Unused IP Address Range

Allocated subnets use:

- **10.10.0.0 – 10.10.0.255** (/24)
- **10.10.1.0 – 10.10.1.127** (/25)
- **10.10.1.128 – 10.10.1.191** (/26)
- **10.10.1.192 – 10.10.1.223** (/27)

The remaining unused addresses are:

- **10.10.1.224 – 10.10.255.255**

This includes:

- The remaining **/27** block from **10.10.1.224 – 10.10.1.255** (32 addresses), and
- All addresses from **10.10.2.0 – 10.10.255.255** within the original **10.10.0.0/16** network.

Final Answer (Summary)

Department	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	**10.10.0.0/24**	**10.10.0.255**	**10.10.0.1 – 10.10.0.254**	**254**
Quality Assurance	**10.10.1.0/25**	**10.10.1.127**	**10.10.1.1 – 10.10.1.126**	**126**
Human Resources	**10.10.1.128/26**	**10.10.1.191**	**10.10.1.129 – 10.10.1.190**	**62**
Executive Management	**10.10.1.192/27**	**10.10.1.223**	**10.10.1.193 – 10.10.1.222**	**30**
Unused Range	**10.10.1.224 – 10.10.255.255**	—	—	Remaining addresses within the **10.10.0.0/16** block

Solution

Department	**Subnet Mask**	**Network Address**	**Valid Host Range**	**Broadcast Address**	**Usable Hosts**
Software Development	/24 (255.255.255.0)	**10.10.0.0**	**10.10.0.1 – 10.10.0.254**	**10.10.0.255**	**254**
Quality Assurance	/25 (255.255.255.128)	**10.10.1.0**	**10.10.1.1 – 10.10.1.126**	**10.10.1.127**	**126**
Human Resources	/26 (255.255.255.192)	**10.10.1.128**	**10.10.1.129 – 10.10.1.190**	**10.10.1.191**	**62**
Executive Management	/27 (255.255.255.224)	**10.10.1.192**	**10.10.1.193 – 10.10.1.222**	**10.10.1.223**	**30**

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1. Network Address and Broadcast Address

Software Development (/24)

- **Network Address:** 10.10.0.0
- **Broadcast Address:** 10.10.0.255

Quality Assurance (/25)

- **Network Address:** 10.10.1.0
- **Broadcast Address:** 10.10.1.127

Human Resources (/26)

- **Network Address:** 10.10.1.128
- **Broadcast Address:** 10.10.1.191

Executive Management (/27)

- **Network Address:** 10.10.1.192
- **Broadcast Address:** 10.10.1.223

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2. Valid Host IP Address Range

Software Development

- **First Host:** 10.10.0.1
- **Last Host:** 10.10.0.254

Quality Assurance

- **First Host:** 10.10.1.1
- **Last Host:** 10.10.1.126

Human Resources

- **First Host:** 10.10.1.129

- **Last Host:** 10.10.1.190

Executive Management

- **First Host:** 10.10.1.193
- **Last Host:** 10.10.1.222

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3. Number of Usable Hosts

The formula used is:

Usable Hosts = 2^(32 − Prefix Length) − 2

Department	Prefix	Total Addresses	Usable Hosts
Software Development	/24	256	**254**
Quality Assurance	/25	128	**126**
Human Resources	/26	64	**62**
Executive Management	/27	32	**30**

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4. Remaining Unused IP Address Range

The allocated subnets occupy:

- **10.10.0.0 – 10.10.0.255** (/24)
- **10.10.1.0 – 10.10.1.127** (/25)
- **10.10.1.128 – 10.10.1.191** (/26)
- **10.10.1.192 – 10.10.1.223** (/27)

The remaining addresses within the **10.10.0.0/16** block are:

- **10.10.1.224 – 10.10.255.255**

This includes:

- The unused portion of the **10.10.1.x** subnet (**10.10.1.224 – 10.10.1.255**), and
- All addresses from **10.10.2.0 – 10.10.255.255**.

Final Answer (Summary)

Department	**Subnet**	**Network Address**	**Valid Host Range**	**Broadcast Address**	**Usable Hosts**
Software Development	/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Remaining Unused IP Address Range

- **Unused IP Range: 10.10.1.224 – 10.10.255.255**

- **Remaining Available Addresses: 10.10.1.224 through 10.10.255.255** within the original **10.10.0.0/16** network block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

1. Convert the IPv6 Address into Compressed Notation

Rules:

- Remove leading zeros in each hexadecimal block.
- Replace one consecutive sequence of all-zero blocks with `::`.

Compressed IPv6 Address

2001:db8:abcd::/64

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2. Expand the Compressed Address Back into Full Notation

Expanding `2001:db8:abcd::` gives:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. Identify the Network Prefix (/64)

A **/64** prefix means:

- The **first 64 bits** represent the network portion.
- The **last 64 bits** represent the interface (host) identifier.

Network Prefix

2001:0db8:abcd:0000::/64

or equivalently,

2001:db8:abcd:0::/64

Network Portion (64 bits):

`2001:0db8:abcd:0000`

Host Portion (64 bits):

`0000:0000:0000:0000`

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4. Calculate the Number of Host Addresses

Host bits = **128 − 64 = 64**

Total possible host addresses:

$$2^{64}$$

$$= 18,446,744,073,709,551,616$$

Therefore,

Total Host Addresses = 18,446,744,073,709,551,616

5. Total Number of Possible Host Addresses Available

Since IPv6 does **not reserve** network and broadcast addresses like IPv4, **all** interface identifiers are available.

Total Possible Addresses:

18,446,744,073,709,551,616 (2^{64})

Final Answer (Summary)

Task	Answer
Given IPv6 Address	**2001:0db8:abcd:0000:0000:0000:0000:0000/64**
Compressed Notation	**2001:db8:abcd::/64**
Expanded Notation	**2001:0db8:abcd:0000:0000:0000:0000:0000/64**
Network Prefix (/64)	**2001:db8:abcd:0::/64**
Network Portion	**2001:0db8:abcd:0000**
Host Portion	**0000:0000:0000:0000**
Host Bits	**64**
Total Possible Host Addresses	**2^{64} = 18,446,744,073,709,551,616**

Solution

1. Convert the IPv6 Address into Compressed Notation

Rules for IPv6 compression:

- Remove leading zeros in each hexadecimal block.
- Replace consecutive groups of 0000 with :: (only once).

Original Address

2001:0db8:abcd:0000:0000:0000:0000:0000

Compressed Address

2001:db8:abcd::

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2. Expand the Compressed Address Back into Full Hexadecimal Representation

Compressed Address:

2001:db8:abcd::

Expanded to its full form:

2001:0db8:abcd:0000:0000:0000:0000:0000

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3. Identify the Network Prefix (/64)

A **/64** prefix means the first **64 bits** represent the network portion, while the remaining **64 bits** represent the interface (host) identifier.

Network Prefix

2001:0db8:abcd:0000::/64

or in full notation:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

- **Network Portion (64 bits):** 2001:0db8:abcd:0000
- **Host (Interface ID) Portion (64 bits):** 0000:0000:0000:0000

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4. Calculate the Total Number of Host Addresses Supported Within the Subnet

Since **64 bits** are available for host addresses:

$$\text{Number of Hosts} = 2^{64}$$

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$$

Total Host Addresses:

18,446,744,073,709,551,616

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5. Total Number of Possible Host Addresses Available Within This Network

IPv6 does **not** reserve network and broadcast addresses as IPv4 does. Therefore, **all** 2^{64}**addresses are usable**.

Total Possible Host Addresses

18,446,744,073,709,551,616 (2^{64})

Final Answer (Summary)

Task	**Answer**
1. Compressed IPv6 Address	**2001:db8:abcd::**
2. Expanded IPv6 Address	**2001:0db8:abcd:0000:0000:0000:0000:0000**
3. Network Prefix (/64)	**2001:0db8:abcd:0000::/64**
4. Total Host Addresses Supported	**2^{64} = 18,446,744,073,709,551,616**
5. Total Possible Host Addresses Available	**18,446,744,073,709,551,616 (all are usable in IPv6)**

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4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

1. Determine the Destination Subnet

For a **/24** subnet mask (255.255.255.0), the first three octets represent the network portion.

Destination IP:

192.168.2.45

Network Address:

192.168.2.0/24

Destination Subnet: 192.168.2.0/24

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2. Matching Routing Table Entry

Compare the destination IP with the routing table:

Routing Table Entry	Match?
192.168.1.0/24	No
192.168.2.0/24	**Yes**
192.168.3.0/24	No

Matching Routing Table Entry: 192.168.2.0/24

3. Next-Hop Network/Interface

Since the destination belongs to **192.168.2.0/24**, the router forwards the packet through the interface (or next hop) connected to that subnet.

Next-Hop Network/Interface:

- **Network: 192.168.2.0/24**
- **Outgoing Interface:** The router interface configured for the **192.168.2.0/24** network (for example, **GigabitEthernet0/1**, depending on the router configuration).

4. Default Route (If No Match Exists)

If the destination IP does not match any routing table entry, the router forwards the packet using the **default route.**

The default route is:

0.0.0.0/0

This route directs unmatched packets to the default gateway or ISP.

Next-Hop Network/Interface

The router forwards the packet through the interface connected to the **192.168.2.0/24** network.

Example:

Destination Network	Outgoing Interface
192.168.2.0/24	**GigabitEthernet0/1** (or the interface configured for this network)

Answer: The packet is forwarded through the interface connected to the **192.168.2.0/24** network (e.g., **GigabitEthernet0/1**).

Note: Since the problem does not provide the actual router configuration, the exact interface name (such as GigabitEthernet0/1) is an example. In practice, the packet is forwarded via whichever interface or next hop is configured for **192.168.2.0/24**.

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4. Default Route

If the destination IP address does **not** match any routing table entry, the router uses the **default route**.

The default route is:

0.0.0.0/0

or

0.0.0.0 0.0.0.0

This route forwards the packet to the router's configured **default gateway (next hop)**.

Final Answer (Summary)

Task	Answer
1. Destination Subnet	**192.168.2.0/24**
2. Matching Routing Table Entry	**192.168.2.0/24**
3. Next-Hop Network/Interface	**Interface or next hop configured for the 192.168.2.0/24 network** (e.g., **GigabitEthernet0/1**)

Task	**Answer**
4. Default Route	**0.0.0.0/0** (used when no routing table entry matches the destination IP)

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

1. Administration LAN (192.168.10.0/26)

- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63
- **Valid Host Range:** 192.168.10.1 – 192.168.10.62
- **Default Gateway:** 192.168.10.1

Suggested IP Addresses

Computer	IP Address
PC 1	192.168.10.2
PC 2	192.168.10.3
PC 3	192.168.10.4

2. Finance LAN (192.168.10.64/26)

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127
- **Valid Host Range:** 192.168.10.65 – 192.168.10.126
- **Default Gateway:** 192.168.10.65

Suggested IP Addresses

Computer	IP Address
PC 1	192.168.10.66
PC 2	192.168.10.67
PC 3	192.168.10.68

LAN	Network Address	Broadcast Address	Valid Host Range	Default Gateway
Administration	**192.168.10.0/26**	**192.168.10.63**	**192.168.10.1 – 192.168.10.62**	**192.168.10.1**
Finance	**192.168.10.64/26**	**192.168.10.127**	**192.168.10.65 – 192.168.10.126**	**192.168.10.65**

Solution

1. Determine the Network Address and Broadcast Address

LAN	Network Address	Broadcast Address
Administration LAN	**192.168.10.0**	**192.168.10.63**
Finance LAN	**192.168.10.64**	**192.168.10.127**

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2. Identify the Valid Host IP Address Range

LAN	Valid Host Range
Administration LAN	**192.168.10.1 – 192.168.10.62**
Finance LAN	**192.168.10.65 – 192.168.10.126**

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3. Suggest Suitable IP Addresses for Three Computers in Each LAN

Administration LAN

Computer	IP Address
PC1	**192.168.10.2**
PC2	**192.168.10.3**
PC3	**192.168.10.4**

(Gateway: 192.168.10.1)

Finance LAN

Computer	IP Address
PC1	**192.168.10.66**
PC2	**192.168.10.67**
PC3	**192.168.10.68**

(Gateway: 192.168.10.65)

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4. Identify the Default Gateway for Each LAN

LAN	Default Gateway
Administration LAN	**192.168.10.1**
Finance LAN	**192.168.10.65**

The default gateway is the router interface that allows hosts in each LAN to communicate with devices outside their local subnet.

Final Answer (Summary)

LAN	Network Address	Host Range	Broadcast Address	Default Gateway
Administration LAN	**192.168.10.0/26**	**192.168.10.1 – 192.168.10.62**	**192.168.10.63**	**192.168.10.1**

LAN	Network Address	Host Range	Broadcast Address	Default Gateway
Finance LAN	**192.168.10.64/26**	**192.168.10.65 – 192.168.10.126**	**192.168.10.127**	**192.168.10.65**

Suggested Computer IP Addresses

LAN	PC1	PC2	PC3
Administration LAN	**192.168.10.2**	**192.168.10.3**	**192.168.10.4**
Finance LAN	**192.168.10.66**	**192.168.10.67**	**192.168.10.68**

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand